Maryland Public School MCAP Proficiency Analysis

Overview

Student proficiency on standardized assessments can vary substantially across subjects, grade levels, and schools. Understanding these patterns alongside school characteristics such as attendance and student demographics can provide additional context for differences in academic performance.

This project analyzes Maryland Comprehensive Assessment Program (MCAP) results from 2023 through 2025 to examine grade-level Math and English Language Arts (ELA) proficiency patterns for grades 3–8. Proficiency is measured as the percentage of tested students scoring at the proficient level or above on the corresponding MCAP assessment. The analysis focused on the following question:

How do Math and ELA MCAP proficiency patterns vary across grades 3–8 in Maryland public schools from 2023–2025, and how are these outcomes associated with attendance and school-level student characteristics?

To address this question, I combined MCAP assessment results with Maryland public school attendance and student-characteristic data. I used Python for data cleaning, exploratory analysis, visualization, and statistical modeling and SQL to integrate the cleaned datasets. The analysis examined proficiency patterns across subject, grade, and year, as well as relationships between proficiency and school characteristics such as chronic absenteeism, economic disadvantage, and student population characteristics.

Data and Methodology

Data Sources

The analysis used publicly available Maryland public school data from 2023 through 2025. Four datasets were used: Math MCAP results, ELA MCAP results, attendance data, and student characteristics data. The MCAP data provided school-level assessment results by grade and subject. The attendance data included measures such as attendance and chronic absenteeism rates, while the student characteristics data included percentages of students participating in Free and Reduced-Price Meals (FARMS) and identified as multilingual learners (ML), students with disabilities (SWD), or gifted and talented (GT).

The analysis was limited to grade-level Math and ELA MCAP assessments for grades 3–8. Course-based Math assessments, such as Algebra I and Geometry, as well as aggregate “All Grades” assessment records, were excluded to maintain a consistent comparison of grade-level Math and ELA results.

Data Preparation and Integration

The datasets were cleaned and standardized in Python before being integrated using SQL. Data preparation included standardizing school and district identifiers, removing aggregate records, aligning variable names across years, and converting reported percentage values into a consistent numeric format.

MCAP results reported as 5% or less or 95% or greater were represented using midpoint values of 2.5% and 97.5%, respectively, while fully suppressed results were treated as missing. These approximations

allowed bounded proficiency results to be retained for analysis while distinguishing them from exact reported values.

The cleaned datasets were integrated at the school and year level using SQL. Some schools had multiple attendance or student-characteristic records corresponding to different school types. To avoid duplicate or incorrect matches, grades 3–5 were matched to Elementary records and grades 6–8 to Middle records when available. If a school had only one contextual record for a given year, that record was used as a fallback, while ambiguous matches were left missing. The resulting analysis dataset contained 22,102 records, with each record representing a unique school, year, grade, and subject combination.

Analysis Approach

Exploratory data analysis was used to examine proficiency patterns across subject, grade, and year and to evaluate relationships between proficiency and school characteristics. Distributions, missing values, and correlations among potential predictors were also examined before developing the regression models.

Ordinary least squares (OLS) regression was used to estimate associations between MCAP proficiency and school characteristics while accounting for differences across subject, grade, and year. A subject-by-grade interaction was included because exploratory analysis indicated that the difference between Math and ELA proficiency varied substantially across grade levels.

A core regression model included chronic absenteeism, FARMS, and SWD, along with subject, grade, year, and the subject-by-grade interaction. An extended model added ML and GT percentages. ML and GT were included in the extended model because these variables contained substantially more missing data, which reduced the number of observations available for the extended model.

Model Evaluation

The regression models were evaluated for linearity, constant variance, residual normality, multicollinearity, and influential observations. Since the data contained repeated observations from the same schools and the residual analysis indicated nonconstant variance, standard errors were clustered by school in the final models. This allowed the uncertainty estimates to account for potential correlation among observations from the same school.

Results

Math and ELA Proficiency Patterns

Across 2023–2025, average ELA proficiency was higher than average Math proficiency at every grade level. As shown in Figure 1, the difference was smaller in the earlier grades but widened substantially in grades 6–8. Average ELA proficiency remained between approximately 42% and 47% across grades, while average Math proficiency declined from approximately 39% in grade 3 to 8% in grade 8.

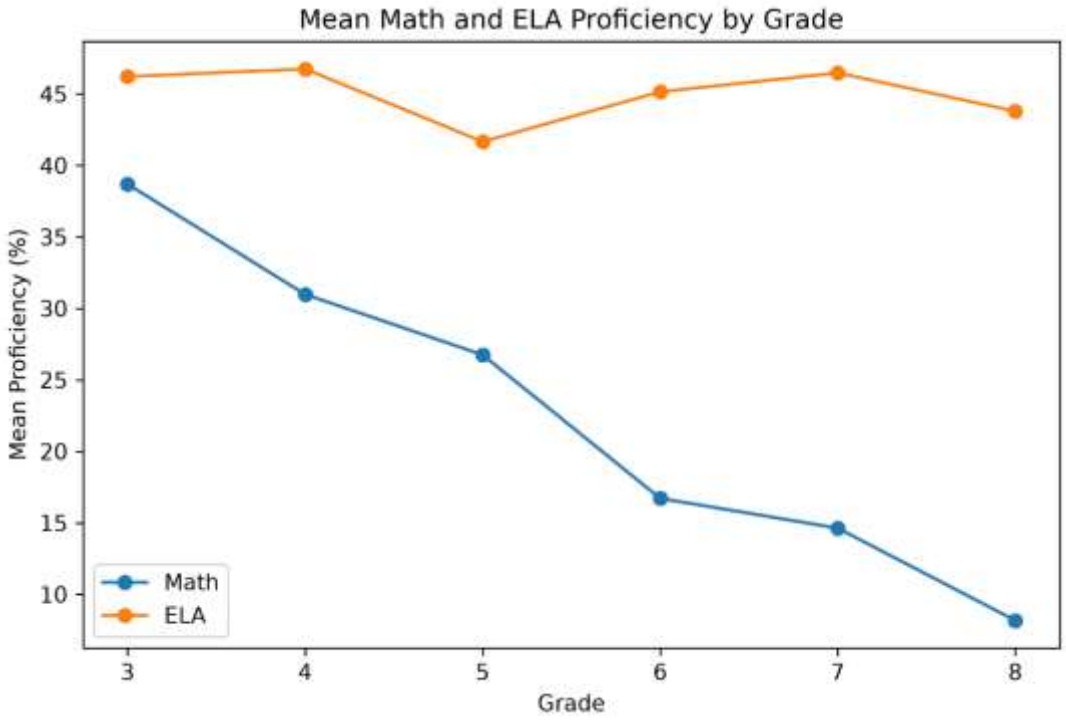


Figure 1. Mean Math and ELA MCAP proficiency by grade, 2023–2025.

The widening difference in the later grades should be interpreted cautiously. Some students in grades 6–8 may take course-based mathematics assessments rather than the corresponding grade-level Math MCAP, so grade-level Math results may not represent all students enrolled in each grade.

Proficiency and School Characteristics

Exploratory analysis showed that chronic absenteeism and FARMS were among the school characteristics most strongly associated with MCAP proficiency. As shown in Figures 2 and 3, schools with higher chronic absenteeism and FARMS percentages generally had lower proficiency rates in both Math and ELA. The overall negative correlations were stronger for ELA than for Math, although proficiency varied considerably among schools with similar characteristics.

These relationships were relatively consistent across 2023–2025, suggesting that the overall patterns were not driven by a single year. However, these findings are descriptive and should not be interpreted as evidence that chronic absenteeism or FARMS participation causes differences in proficiency. Regression models were used to examine whether the associations remained after accounting for subject, grade, year, and other school characteristics.

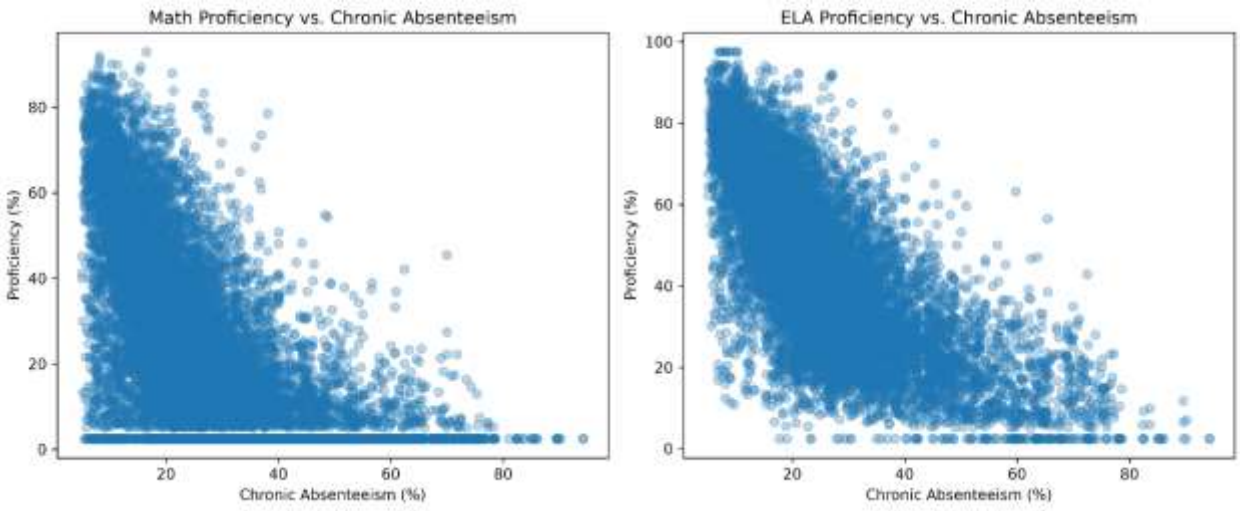


Figure 2. Math and ELA MCAP proficiency by chronic absenteeism rate.

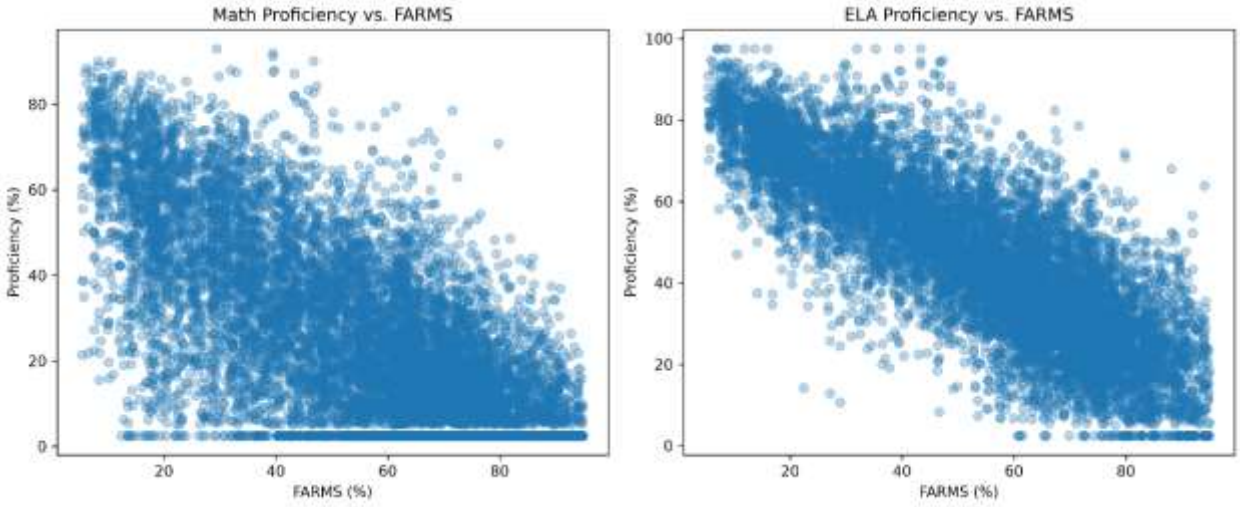


Figure 3. Math and ELA MCAP proficiency by FARMS percentage.

Regression Results

The core regression model included 19,530 observations and explained approximately 75.7% of the variation in MCAP proficiency. After accounting for subject, grade, year, and other school characteristics, both chronic absenteeism and FARMS remained negatively associated with proficiency. A one-percentage-point increase in chronic absenteeism was associated with an estimated 0.28-percentage-point decrease in proficiency, holding the other variables constant. Similarly, a one-percentage-point increase in FARMS was associated with an estimated 0.59-percentage-point decrease in proficiency. Both associations were statistically significant.

The core model also estimated a positive association between SWD percentage and proficiency, with a one-percentage-point increase in SWD associated with a 0.19-percentage-point increase in predicted proficiency, holding the other variables constant. However, this finding should be interpreted cautiously

because the estimated association was sensitive to the sample used in the analysis. The regression results also supported the earlier finding that the Math–ELA proficiency difference varied substantially across grade levels.

The extended model included ML and GT percentages but used a smaller sample of 7,509 observations because of missing data for these variables. The model explained approximately 80.0% of the variation in proficiency. Chronic absenteeism and FARMS remained negatively associated with proficiency, with estimates similar to those in the core model. In contrast, the SWD coefficient became slightly negative and was no longer statistically significant. This change suggests that the estimated association between SWD and proficiency was less consistent across the models.

In the extended model, higher ML percentages were significantly associated with lower proficiency, while the positive estimated association for GT percentage was not statistically significant at the 0.05 level. Because the extended model was estimated using fewer observations, differences between the models may reflect both the additional predictors and which observations were included.

Conclusion

This analysis found substantial differences in MCAP proficiency across subjects and grade levels. Average ELA proficiency was higher than Math proficiency across grades 3–8, with the difference widening considerably in the later grades. Higher chronic absenteeism and FARMS percentages were also consistently associated with lower proficiency in both the exploratory analysis and regression models, even after accounting for subject, grade, year, and other school characteristics.

Several limitations should be considered when interpreting these results. Because the analysis used observational, school-level data, the relationships identified should not be interpreted as causal or as relationships at the individual student level. Some MCAP proficiency values were approximated using midpoints of reported ranges, while fully suppressed values were treated as missing. Additionally, grade-level Math results for grades 6–8 may not represent all students enrolled in those grades because some students take course-based assessments instead. The extended regression model also used a smaller sample due to substantial missing data for ML and GT percentages.

Future analysis could incorporate additional years of data and examine whether the observed relationships vary across districts or geographic areas. Including additional school or community characteristics could also provide further context for differences in proficiency.